Lithium Battery Safety System (BSS)

Kretto

KRETTO — Power Solution —

Professional Provider of Intelligent Operation and Maintenance Solutions for Power Lithium Batteries

Online-to-Offline Model

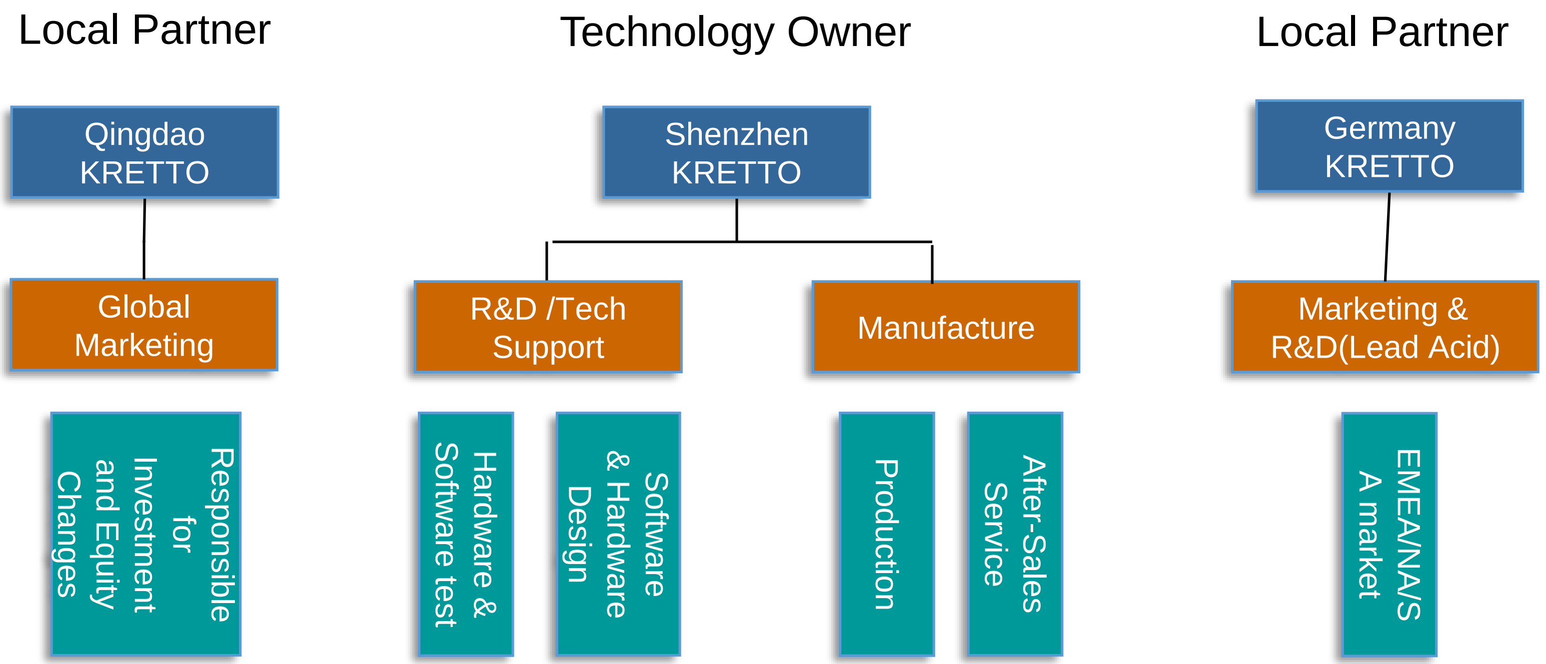
By continuously tracking and analyzing real-time battery data online, the system can quickly identify anomalies in battery performance. Combined with offline edge computing and rapid detection capabilities, it enables early identification of abnormally degrading batteries and provides maintenance recommendations—effectively eliminating potential thermal runaway risks at an early stage.

Multi-Dimensional Service Platform

Track and analyze the operational data of new energy vehicles, light electric vehicles, and energy storage stations to deliver comprehensive services including battery performance evaluation, safety assessment, and maintenance support—ultimately building an industry-wide service platform for lithium battery operation and maintenance.

Organizational structure

KRETTO has senior technical personnel with long-term working experience and theoretical knowledge in the fields of batteries, electrical equipment, IT and artificial intelligence, and has formed a network with overseas partners in various countries with global business capabilities.



Key experts

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Dr. Jiang Jiuchun phD

A leading figure in the power battery management industry, formerly a professor and doctoral advisor at Beijing Jiaotong University, currently serving as Chief Scientist at the Shenzhen Automotive Research Institute, Beijing Institute of Technology.
Recipient of the National High-Level Talent Special Support Program (Ten Thousand Talents Plan),
China Automotive Industry Science and Technology Award (First Prize),
China Machinery Industry Science and Technology Award (Special Prize),
National Science and Technology Progress Award (Second Prize).

Dr. Yan Huiwen phD

Ph.D. in Power Electronics from Tsinghua University.
Shenzhen Leading Talent.
Expert Reviewer for New Energy Projects.
EMBA from China Europe International Business School (CEIBS).
Former R&D Project Manager at Emerson
Chief Engineer at Kstar
Holds over 20 patents.
Currently an expert committee member and reviewer for the China Power Supply Society.

Dr. Wu Zhiqiang phD

Senior Engineer, expert registered with the Ministry of Science and Technology, and special expert with the China Association for the Promotion of International Trade.
Long-term engaged in electric vehicle technology research, pioneered the “Three-Network Integration” technology framework and its application. Focuses on battery analysis technology and applications involving mechanism and data fusion.
Former expert member of the Electric Vehicle Standardization Committee in Guangdong Province and Shanghai.
Has led and participated in over 10 national key science and technology projects, and holds more than 10 authorized invention patents.

Key experts

Dr. Jiang Yan phD

Senior Engineer with a Doctoral Degree; Visiting Scholar at the University of California, San Diego from 2018 to 2019.
Involved in 10 projects, including National Natural Science Foundation and National Key R&D Program projects.
Published 14 SCI papers and holds 4 authorized invention patents.
Mainly responsible for the development of battery performance evaluation strategies.

Dr. Gao Yang phD

Senior Engineer, responsible for core algorithm development, with a Doctoral Degree.
Involved in National Natural Science Foundation and National Key R&D Program projects.
Published over 10 SCI and EI indexed papers and holds 4 authorized invention patents.
Primarily responsible for the development of battery safety warning strategies.

Dr. Wei Shaoyuan phD

Senior Engineer with a Doctoral Degree; Visiting Scholar at Chalmers University of Technology, Sweden, from 2019 to 2020.
Involved in nearly 10 National Key R&D Program projects and provincial/municipal-level projects.
Published over 10 SCI and EI journal papers and holds 2 authorized invention patents.
Primarily responsible for capacity optimization and energy management strategies.

2 Background

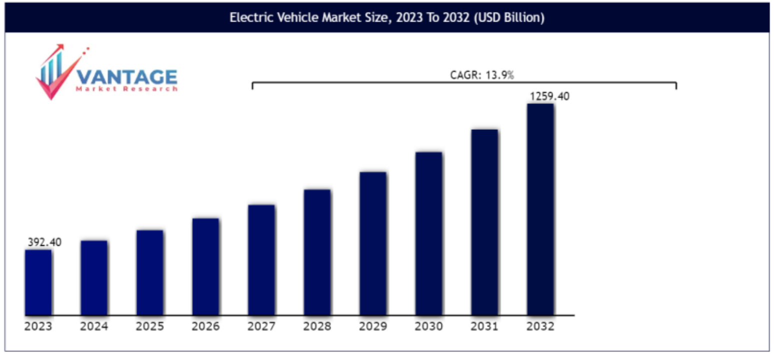
Marketing environment

Driven by the global shift toward eco-friendly energy and electrification, the automation of application products is evolving rapidly. Alongside the transformation of industrial structures, the global market for lithium-ion battery–related industries is experiencing rapid growth.

EV Market

*source : Vantage Market Research (Nov.,2022)

The global electric vehicle (EV) market is valued at $392.4 billion as of 2023 and is projected to grow at a compound annual growth rate (CAGR) of 13.9% from 2024 to 2032, reaching $1,259.4 billion by 2032.

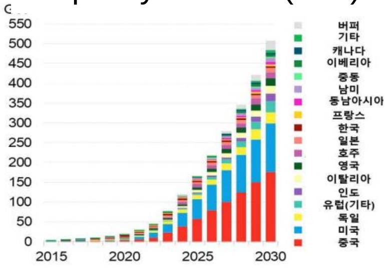


ESS Market

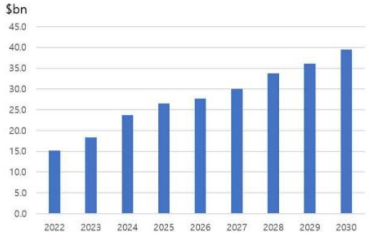
*source : BNEF (Nov.,2023)

The global energy storage system (ESS) market is expected to grow from $15.2 billion in 2022 to $39.5 billion by 2030. Over the same period, the total installed ESS capacity is projected to rise from 43.8 GW to 508 GW.

World ESS Installed Capacity Outlook (GW)



World ESS grows size



LIB Market

*source : CM Insights (Oct.,2022), McKinsey (Jan.,2023)

The global lithium battery market was valued at $48.8 billion in 2022 and is projected to grow at a compound annual growth rate (CAGR) of 18.5%, reaching $184.1 billion by 2030. In terms of global demand, total capacity is expected to rise from 700 GWh in 2022 to 4,700 GWh by 2030.

2 Background

Lithium battery safety issues

With the rapid expansion and large-scale deployment of new energy vehicles and lithium battery energy storage systems, safety incidents involving lithium batteries have become increasingly frequent—emerging as a critical challenge for the industry. However, battery manufacturers, OEMs, and system operators often lack the professional capabilities to accurately assess the condition of batteries in use.

Batteries form an endless web of energy, yet they also represent countless double-edged swords hanging over modern society—powerful, but potentially dangerous if left unchecked.

Battery fires occur frequently

Fire incidents primarily involve new energy vehicles, energy storage stations, and light electric vehicles.

Large direct economic losses

The cumulative economic losses caused by battery-related fire incidents have exceeded USD 1 billion

Recall growth

The primary cause lies in safety risks associated with power batteries, affecting vehicle brands across high, mid, and low-end segments, and involving both passenger and commercial vehicles.

As of August 2021, a total of 10 manufacturers had issued recalls due to potential thermal runaway risks in battery packs, affecting 60,940 vehicles.

2 Background

Lithium battery maintenance management limitations

01 Manufacturing technological level

The reliability of batteries and their system components is a key factor in ensuring overall battery system safety. Incomplete battery testing, inadequate performance and safety assessments, and a lack of regular maintenance significantly increase the risk of battery-related safety incidents.

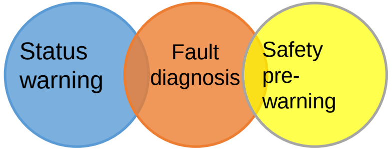
02 Abnormal state of normal aging caused by use

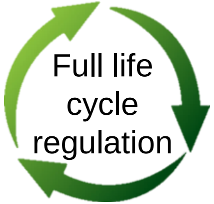
As lithium batteries are used over time, aging and consistency issues gradually become more pronounced, significantly reducing the overall utilization efficiency and service life of the battery system.

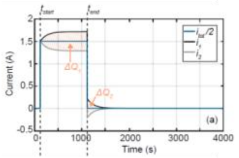
03 After sales service and aftermarket

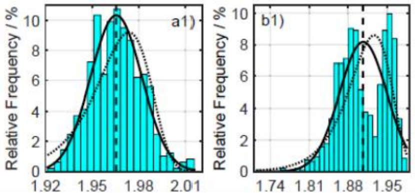
The daily maintenance, safety management, and asset evaluation of lithium batteries urgently require professional tools and service support.

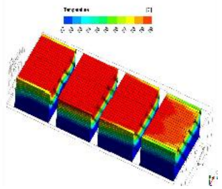
BSS is born of three objective needs

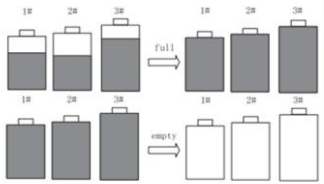












- Cell failure
- Cell BMS failure
- Battery housing broken and seal failure
- Insulation failure
- Connection failure

Lithium batteries are inherently dangerous

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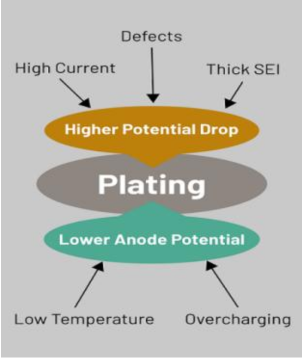
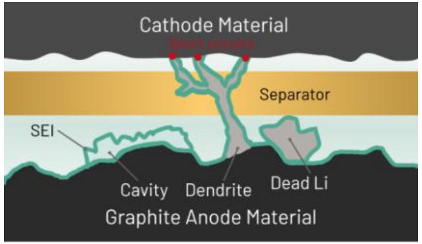
During the charging process of a lithium battery, lithium ions migrate from the anode material into the layered structure of the cathode material. Under certain conditions, some of these ions deposit as metallic lithium in the cathode region, forming dendritic lithium structures. Internal and external battery anomalies can accelerate this plating process. In severe cases, the rapid growth of lithium dendrites may penetrate the battery separator, causing internal short circuits. This can trigger thermal runaway, potentially leading to serious safety incidents such as fires.

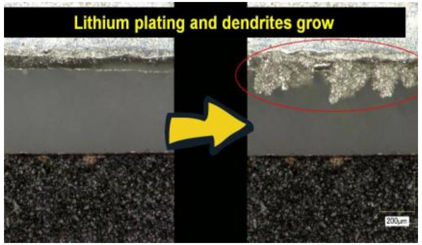
Lithium Plating & Dendrite Growth

- Short circuit
- Large current charging
- Overcharge overdischarge
- Structural damage
- External overheating
- Crash

- ✓ Electrochemical reaction
- ✓ Internal short circuit
- ✓ Disintegrate
- ✓ Electrode-electrolyte reaction







Leakage

Explode

Smoke

Thermal Runaway

Fire

Air

Limitations of existing BMS

The Battery Management System (BMS) is designed to manage and protect batteries while providing essential control information to ensure the safe and efficient operation of application products such as electric vehicles and energy storage systems. However, conventional BMS technology lacks the accuracy and reliability needed to keep up with the rapid evolution of lithium battery technologies. Moreover, traditional BMS can only detect issues after they occur, without the ability to predict potential safety risks—significantly limiting its effectiveness.

Main function of existing BMS

Monitor the battery system status
SoC analysis
SoH analysis
Fault diagnosis

Lithium battery cell management
Cell balancing

Battery control
Over charge/discharge control
Over current control

\+ **BSS**
SoC, SoH reliability
Fault diagnosis accuracy
Thermal runaway early warning

= **Advanced BMS**

Existing BMS technologies offer only 40% to 60% prediction accuracy. For thermal runaway—often leading to fires or explosions—warning times are typically limited to just a few seconds to five minutes.

(No alarm before accident, and battery was smoking)

Battery failures and fires pose serious risks for EV users and ESS operators. For major incidents like explosions, timely countermeasures are critical—especially when early warnings provide sufficient lead time.

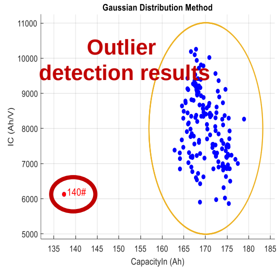
(Five seconds after smoke, fire)

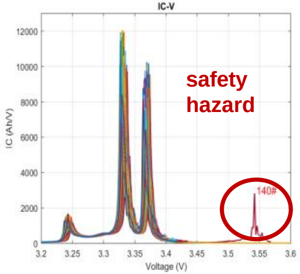
3 Solution

Accumulate large amounts of battery data

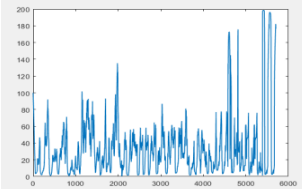
For a long time, Qingdao KRETTO has been systematically conducting battery testing under the guidance of experts, accumulating a vast amount of battery performance data. This comprehensive battery database forms the backbone of KRETTO’s AI-powered BSS products and services, giving them a strong competitive edge and enabling clear product differentiation in the market

- Testing time: more than 10 years
- Testing place: China National Laboratory, participated in by leading national-level scientists from China
- Testing objects
  - Application product group: EV (passenger car, commercial vehicle, low speed electric vehicle, etc.), ESS
  - Battery type: prismatic, cylindrical, pouch cell
- Test quantity: More than 700,000 test vehicles (more than 700,000 battery packs)
- Test results: more than 6,000 kinds of batteries operation data

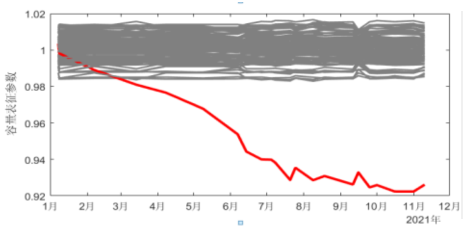


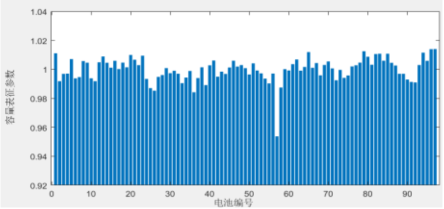


✓ Identify lithium plating in batteries and provide early warnings of possible thermal runaway events..



✓Conduct periodic assessments of battery condition, identify faults in offline batteries, and carry out timely replacement and maintenance.





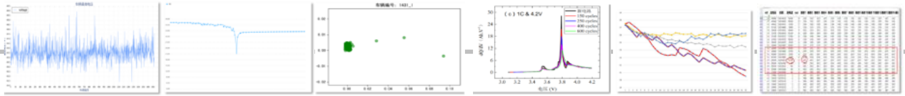
✓ Early detection of rapid capacity degradation allows for the proactive replacement of abnormal batteries, thereby prolonging the service life of the entire battery system.

3 Solution

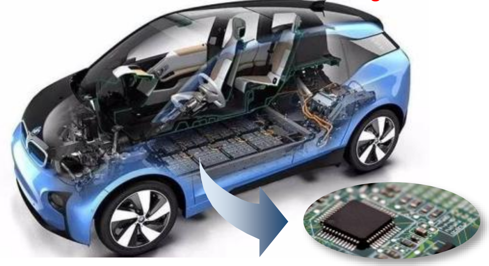
Artificial Intelligence (AI) based solution

KRETTO's BSS is built upon an extensive database of battery operation analytics and incorporates advanced artificial intelligence algorithms that account for various electrochemical mechanisms of lithium-ion batteries. The solution developed by KRETTO represents a world-leading innovation, enhancing traditional battery management systems with cutting-edge technologies to maximize battery utilization efficiency and ensure the highest level of safety for customers.

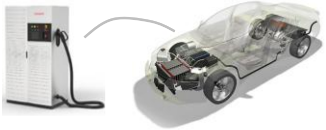
Battery algorithm model



With 8 comprehensive battery evaluation models, more than 12 key analytical indicators, and multi-timescale algorithms, the platform ensures complete (100%) coverage of all battery assessment dimensions.



Edge computing module



Detecting equipment

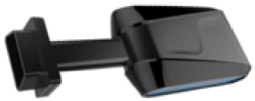
Big data monitoring platform

A dedicated platform that integrates analytics and service applications, offering open service capabilities to support diverse application scenarios. It enables full life cycle management for millions of battery systems and currently manages over 100,000 battery systems in total, ensuring stable and reliable operation.

3 Solution

BSS can be used without changing the original system

(Example)
EV battery detection



(BSS Data Collector)

OBD Interface Location in EVs for Easy Installation of the BSS Data Collector

What is OBD (**O**n **B**oard **D**iagnostics) ?
Diagnostic and monitoring devices connected to vehicle or equipment control systems

Users can connect the terminal (BSS Data Collector) directly to the vehicle’s OBD interface. As shown in the illustration, the OBD port is typically located behind a panel at position A or B for easy access.

The vehicle’s OBD port is designed to prevent incorrect insertion of the terminal, and the connector is usually white or black in color.

The terminal should be installed near the vehicle‘s OBD port and must not be enclosed by any metal surfaces.

Once the installation location is selected, the terminal can be securely fixed using the 3M double-sided adhesive on its back, or alternatively, it can be fastened with a cable tie.

Simple user interface

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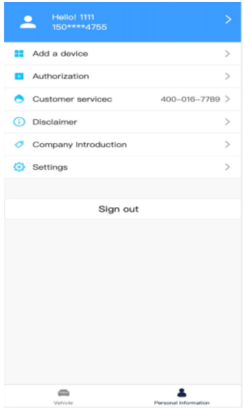
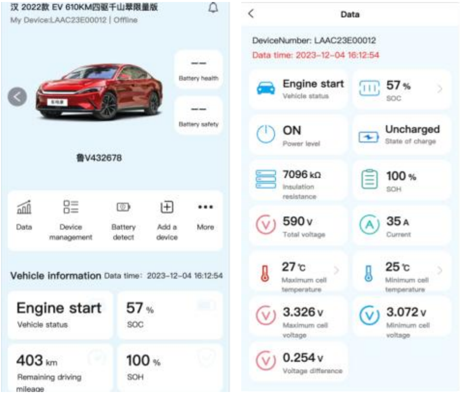
(Example)
EV battery detection

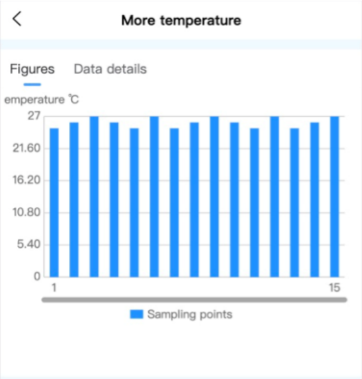
(User log in)

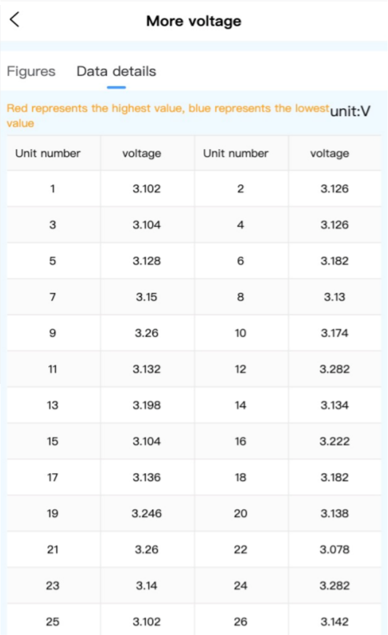


(Equipment basic information)



(Equipment detailed information)





Unit number	voltage	Unit number	voltage
1	3.102	2	3.126
3	3.104	4	3.126
5	3.128	6	3.182
7	3.15	8	3.13
9	3.26	10	3.174
11	3.132	12	3.282
13	3.198	14	3.134
15	3.104	16	3.222
17	3.136	18	3.182
19	3.246	20	3.138
21	3.26	22	3.078
23	3.14	24	3.282
25	3.102	26	3.142

- A push-based warning system is provided to deliver real-time alerts and notifications to vehicle users.
- Personal information of vehicle users is protected through data encryption to ensure privacy and security.

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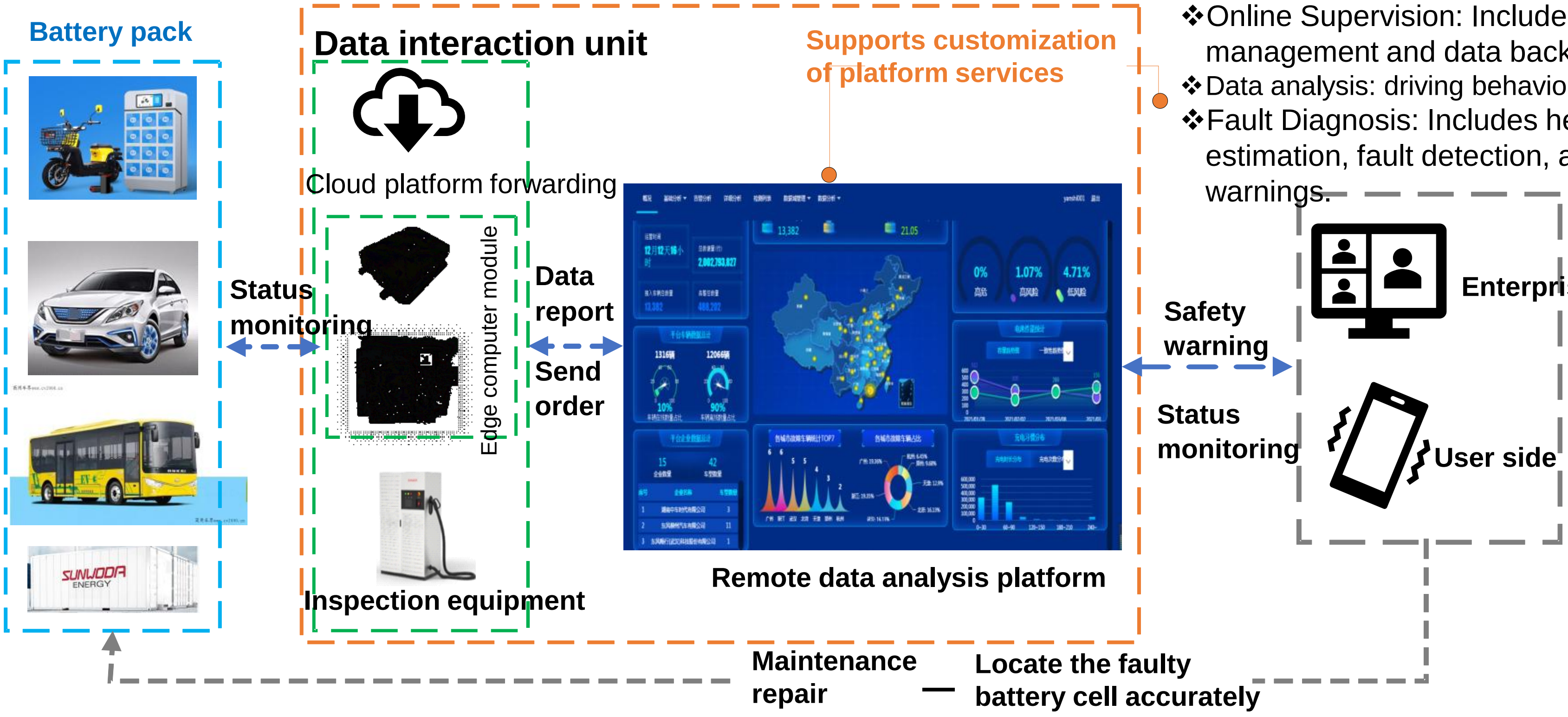
"Vehicle-Ground-Cloud" collaborative solution

- ❖Battery Management: Includes order registration and asset management.
- ❖Online Supervision: Includes pattern management and data backtracking.
- ❖Data analysis: driving behavior/habit
- ❖Fault Diagnosis: Includes health state estimation, fault detection, and safety warnings.



4 Product features

Product architecture

Early warning system

- ✓ Capable of detecting potential faults or abnormal behavior in the battery pack 12 to 24 hours in advance, enabling early intervention and risk prevention.
- ✓ Utilizes an exclusive database covering over 6,000 types of battery BMS.
- ✓ Capable of detecting 20 different factors that affect battery safety.

Simple and versatile product features

- ✓ AI-based software can analyze new and unknown BMS types and seamlessly integrate them into existing systems.
- ✓ LCO, NCM, LFP, all types of batteries.
- ✓ Customer solutions or applications utilize commercial 4G/5G networks to transmit user-related information.

Central server system

Local Edge Computer / Data Collector

4 Product features

Product architecture

Cloud side

New Energy Vehicle Safety Warning and Analysis Platform

Intelligent Operation and Maintenance Platform for Energy Storage

Ground side

Diagnostic unit

Intelligent data terminal

Battery balancing equipment

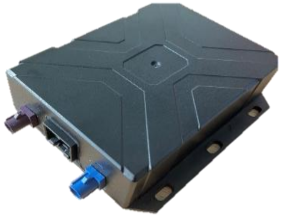
Charging control gateway

120kW detecting system

60kW detecting system

Vehicle side

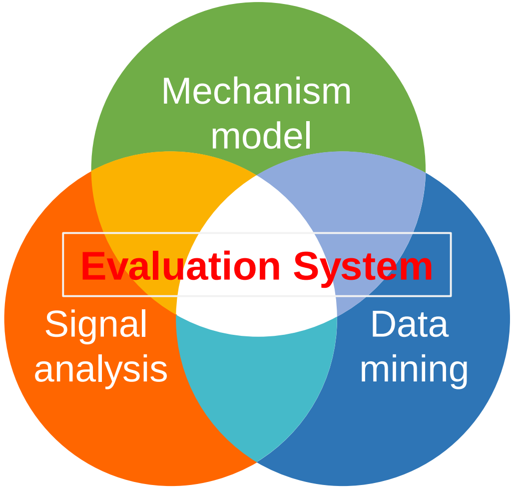
Wireless communication module



T-BOX

4 Product features

Battery performance and safety evaluation system



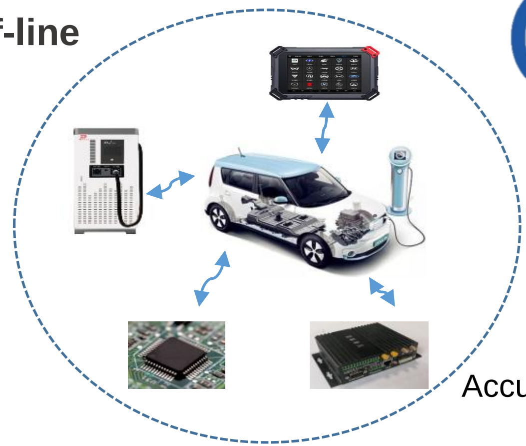
- Identify security risks in advance
- Precise fault location
- Battery life extension
- Energy efficiency optimization operation

On-line

Battery side big data as support
(Establishment of medical records)

Abnormal aging Long-term: monthly	Microshort circuit Middle stage: Daily	Severe internal short-circuit Short-term: per hour/minute

Off-line

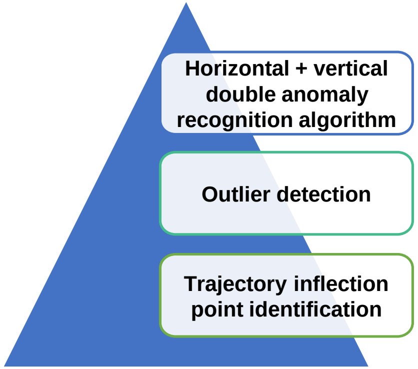


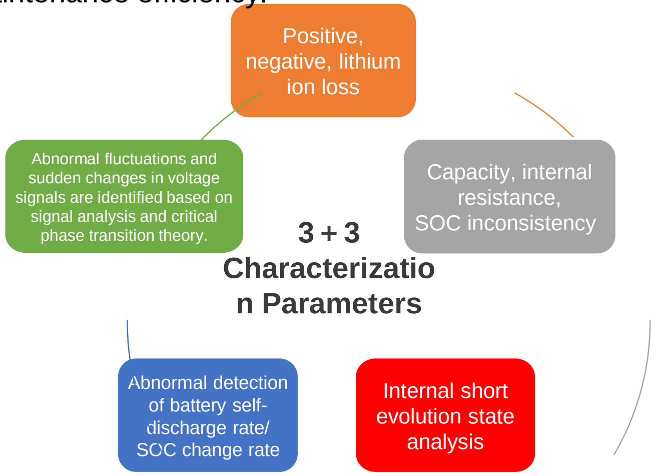
Accurate detection even in the absence of cloud data
(via regular check-ups, designated inspections, or targeted maintenance)

Battery performance and safety evaluation system

A multi-timescale battery performance and safety evaluation system has been established using a combination of mechanism modeling, data mining, and signal analysis.

- ✓ Comprising 3 main categories and 12 indicators, the system performs a multi-level comprehensive evaluation of battery faults. The algorithm is applicable to all operating conditions and achieves a battery anomaly detection rate of over 95%.
- ✓ Early identification of battery safety risks to improve inspection and maintenance efficiency.





An integrated battery capacity estimation framework combining original parameter identification, machine learning, and curve splicing.

- ✓ Overcomes challenges in estimating vehicle battery capacity under various environmental conditions, including low data quality and incomplete charge/discharge cycles.
- ✓ The BSS battery capacity estimation error is less than 3%.

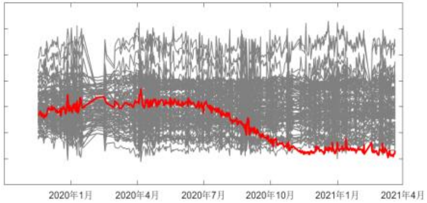
4 Product features

Cloud big data monitoring services

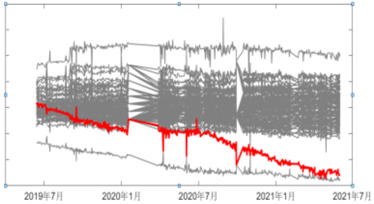
All battery-related information—including status evaluation, defect detection, and early safety warnings—will be provided to customers in real time via a cloud-based big data monitoring platform. Core functions include early warnings of fire or explosion risks caused by thermal runaway, diagnostics of abnormal battery aging, comprehensive evaluations of battery cycling behavior, and predictive insights for warranty management.

Cloud big data monitoring platform

Thermal Runaway early warning



- ✓ In July 2020, a micro short circuit was detected, and in April 2021, the vehicle experienced a thermal runaway event.



- ✓ In July 2020, a micro short circuit was identified, and in June 2021, the vehicle experienced a thermal runaway incident.

4 Product features

Cloud big data monitoring services

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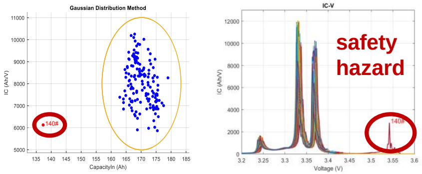
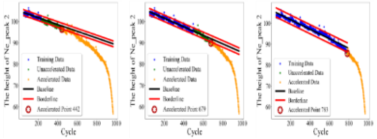
Abnormal aging diagnosis

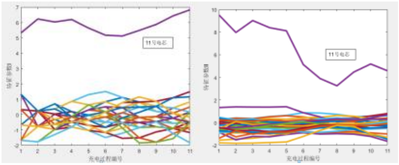


Outlier detection results

✓ Lithium Plating diagnosed

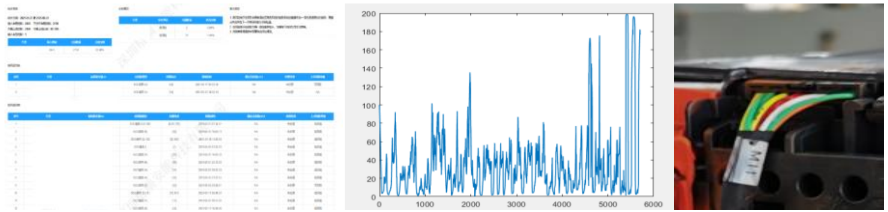


✓ Identify capacity dives

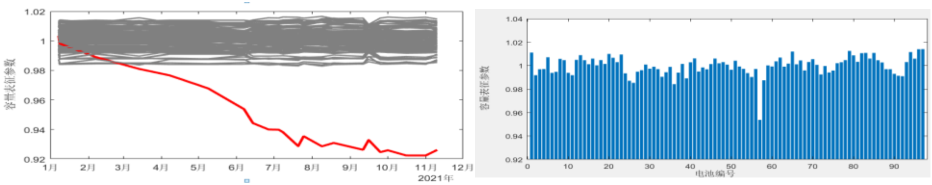


✓ #11 Abnormal evolution observed in lithium-ion loss and negative electrode material degradation.

Periodic comprehensive battery evaluation and predictive maintenance.



Periodically evaluate battery status, identify faults offline, and provide guidance for battery replacement and maintenance.



✓ By detecting rapid battery capacity decay, abnormal batteries can be proactively replaced to extend the overall service life of the battery system.

High level of accuracy and reliability

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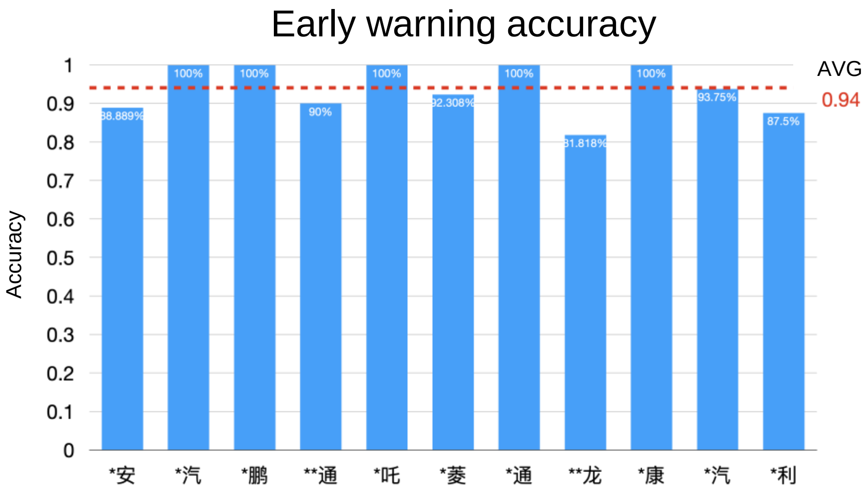
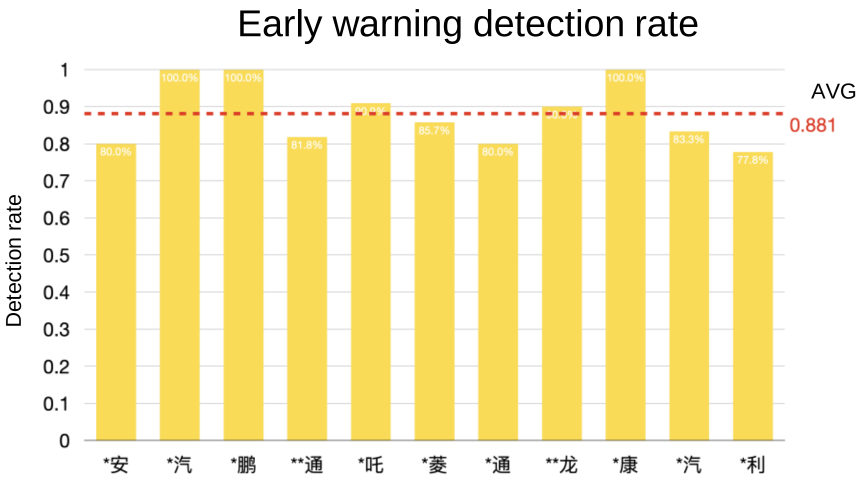
Based on BSS's extensive empirical testing on operating vehicles, the system achieves a battery safety warning accuracy of **over 94%,** and a thermal runaway prediction accuracy exceeding 88%. The average early warning lead time is **11.4 hours**—significantly outperforming the current industry standard of **less than 5 minutes**. As the database continues to grow, encompassing a wider range of EV and ESS battery types, the system's accuracy and detection sensitivity are expected to further improve.





High level of accuracy and reliability

KRETTO utilizes its BSS system to estimate the battery capacity of electric vehicles, confirming the vehicle's actual available capacity through intermittent testing. The resulting error rate is less than 3%, representing a leading level in the industry. As the database continues to expand with data from various types of EV and ESS batteries, estimation accuracy will further improve, leading to even higher reliability.

Test vehicle	Total mileage	Full charge capacity	Interval test available capacity	Result deviation
A8****	22433	130.68Ah	128.6Ah	-1.59%
D*****	55595	115.8Ah	116.9Ah	0.95%
G9*****	89940	110.00Ah	107.1Ah	-2.63%
G1*****	40186	121.8Ah	120.98Ah	-0.67%
E******	97368	104.8Ah	107.32Ah	2.40%
K*****	28901	139.2Ah	141.8Ah	1.87%
JA8****	32023	137.7Ah	140.9Ah	2.32%






BSS profit model

BSS Diagnostics Service

The BSS Data Collector is installed on the battery side of applications such as EVs or ESS systems. It gathers data from the BMS and transmits it to the cloud, where the AI-powered BSS software analyzes battery status and safety conditions. The analysis results are then delivered to customers in real time. Revenue is generated through service usage fees paid by customers during the operation period.

Advanced BMS (BSS Module) supply

The BSS Module replaces the original BMS installed in EV or ESS battery packs. It is co-designed with application product developers or battery pack manufacturers, and is either produced directly or procured by OEMs. Revenue is generated through BSS Module sales and BMS design service fees.

Flexible expansion of business models

Insurance Companies:
Partner to reduce fire risks through BSS diagnostic services, helping insurers lower claims and enhance risk control.
Auto Maintenance Centers:
Enable smart EV diagnostics, battery evaluation, and predictive maintenance management.
IoT Platform Operators:
Integrate BSS to enhance IoT services for EVs, UPS, and ESS with real-time battery insights.
Government/Public Authorities:
Collaborate to build EV and ESS fire monitoring and early warning systems, improving public safety.

B2C Model

Direct User Service Model:
Provide real-time battery monitoring and safety early warning services. Users can obtain real-time updates on the battery safety status by subscribing or purchasing the service.
Provide warnings and maintenance suggestions regarding the battery status to individual users through mobile phone apps or computer terminals.
Connect directly to the vehicle owner's OBD port through on-board equipment to provide simple and easy-to-use safety monitoring.

Insurance Cooperation Model:
Cooperate with insurance companies to reduce users' fire risk insurance premiums. The BSS system reduces the risk of battery-related accidents through precise battery monitoring, thereby enabling users to obtain premium discounts.

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B2B Model

OEM and Module Supply:

Cooperate directly with battery manufacturers or automobile manufacturers to use the BSS module as a supplement to the original BMS (Battery Management System). Customize the BSS module through OEM cooperation to meet the needs of specific applications.

The revenue sources of the cooperation include the sales of BSS modules, design customization and related technical service fees.

Maintenance and Inspection Cooperation:

Cooperate with automotive repair centers and charging stations to establish intelligent battery inspection and maintenance service stations. Provide services including comprehensive safety evaluation, battery maintenance and status diagnosis.

This cooperation method can enhance the customer service capabilities of enterprises and promote the popularization of the BSS system in a wider range of application scenarios.

IoT Platform and Big Data Monitoring Cooperation:

Cooperate with Internet of Things (IoT) platform operators to integrate the data of the BSS system into the IoT ecosystem and expand to more IoT application scenarios, such as electric vehicles, ESS and data centers.

Provide data platform support to realize real-time upload of battery status data, cloud analysis and remote management, and support the full life cycle monitoring of large-scale battery systems.

Government and Public Institution Collaboration Model:

Cooperate with governments and public safety agencies to establish fire early warning and monitoring systems for public infrastructure and energy storage systems. Support the establishment of a public safety network to prevent and respond to potential battery safety accidents.

6 Business strategy

After-Sales Service

Product training

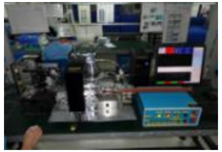
Provide training tailored to the user's situation and actual needs, covering not only product installation and usage, but also other relevant topics as required.

Consultation service

7×24-hour technical support and consulting services available via telephone hotline or WeChat.

Technical support

During the warranty period, free maintenance and technical support are provided for terminals not damaged by human factors.



Regular visit

By regularly inspecting data transmitted from the OEM platform terminal, potential issues can be identified early to enhance service quality.

Service upgrade

Provide system expansion, version upgrades, and feature updates to ensure stable and reliable system operation under normal conditions within the defined scope of requirements.

Product warranty policy

During the warranty period, if issues cannot be resolved remotely, on-site technical support can be provided based on the actual situation.

These characteristics ensure the leading position of the BSS system in the global battery management fiel

1. AI-driven Battery Monitoring

Utilizing artificial intelligence and big data, BSS can predict battery faults 12-24 hours in advance, which is more accurate than traditional BMS, and the early warning accuracy rate exceeds 94%.

2. Vast Data Support

The system has accumulated operating data of more than 700,000 battery packs and 6,000 types of battery cells, enabling more comprehensive state assessment and real-time monitoring. The early warning time is 11 hours earlier than the industry standard.

3. Full Life Cycle Monitoring

Provide full-process monitoring from manufacturing to usage, detect battery status through multi-time-scale algorithms, extend lifespan and ensure safety.

4. Edge Computing and Cloud Integration

Combining edge computing and cloud monitoring platforms, it supports large-scale real-time data transmission and can be installed without replacing the existing system.

5. Strong Adaptability and Expandability

Suitable for various scenarios such as electric vehicles and energy storage systems, and supports cooperation with multiple parties such as IoT platforms and insurance companies, with high flexibility and expandability.

Thank you!

Contact

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